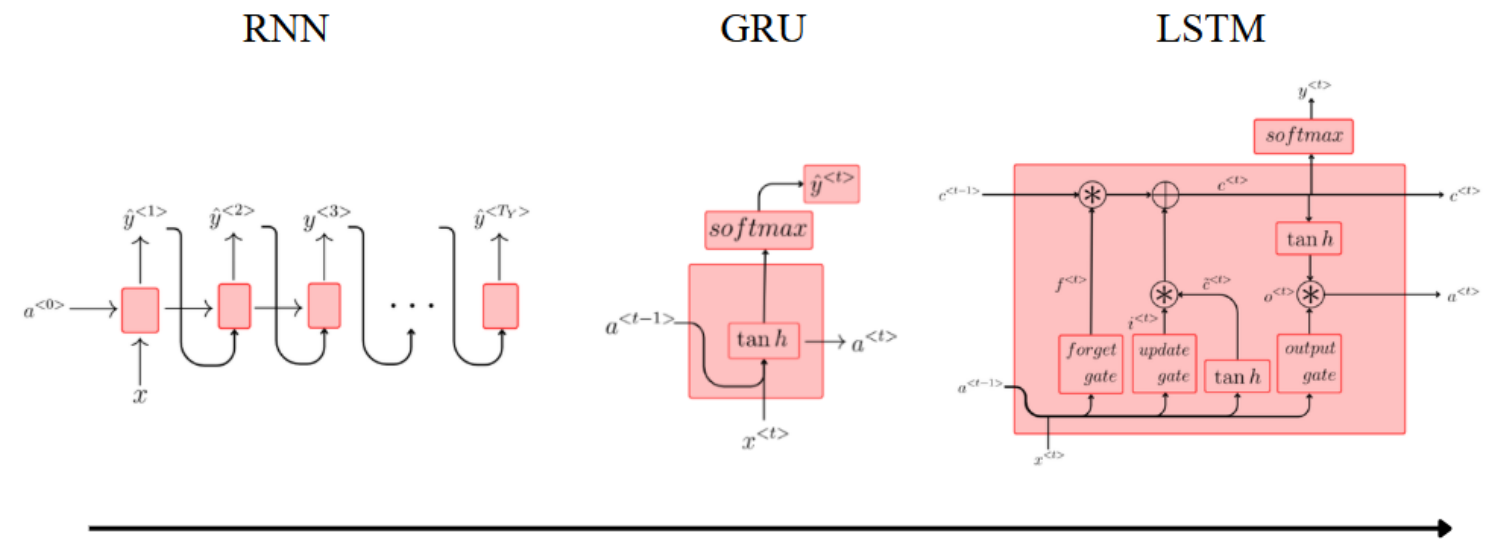

Machine Learning 7

Motivation



Note: Complexity goes up, yet the models are still sequential. i.e., they ingest data at a single token/unit at a time.

Note: We would ideally want a structure which gives us some of the parallel nature of a CNN, as well as some of the sequential-ness of a RNN.

The transformer network should be understood in 3 major components,

- attention
- self-attention
- multi-head attention

Transformer Network

Self-Attention

$A(q, k, v)$ = "attention-based vector representation of a word"

where we calculate for each word $A^{<1>}, A^{<2>}, \dots, A^{<5>}$.

RNN Attention

$$\alpha^{<t, t'>} = \frac{\exp(e^{<t, t'>})}{\sum_{t'=1}^{T_x} \exp(e^{<t, t'>})}$$

Transformer Attention

$$A(q, k, v) = \sum_i \frac{\exp(q \cdot k^{<i>})}{\sum_j \exp(q \cdot k^{<j>})} v^{<i>}$$

Example:

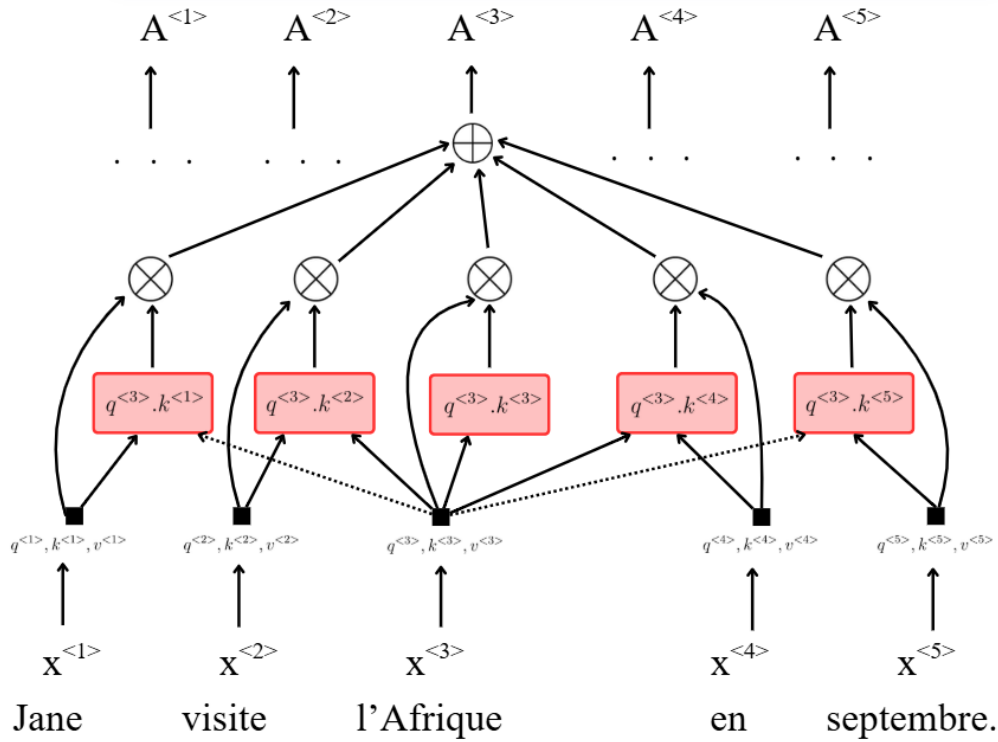
for the 5 word example, we can have different attention values,

$$A^{<3>} = A(q^{<3>}, K, V)$$

where

Query (Q)	Key (K)	Value (V)
$q^{<1>}$	$k^{<1>}$	$v^{<1>}$
$q^{<2>}$	$k^{<2>}$	$v^{<2>}$
$q^{<3>}$	$k^{<3>}$	$v^{<3>}$
$q^{<4>}$	$k^{<4>}$	$v^{<4>}$
$q^{<5>}$	$k^{<5>}$	$v^{<5>}$

The complete infrastructure for one node will look like,



we have,

- $q^{<3>} = w^Q . x^{<3>}$
- $k^{<3>} = w^K . x^{<3>}$
- $v^{<3>} = w^V . x^{<3>}$

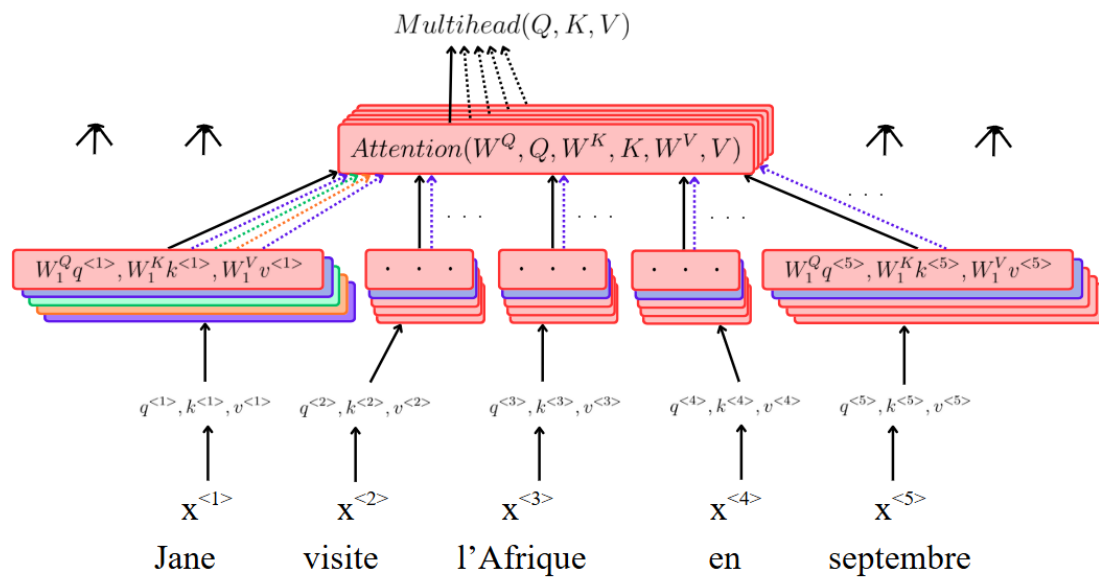
$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

Multi-head Attention

- Conceptually is just a big for loop over the self-attention process.
- Each time you calculate self-attention for a sequence is called a "head".

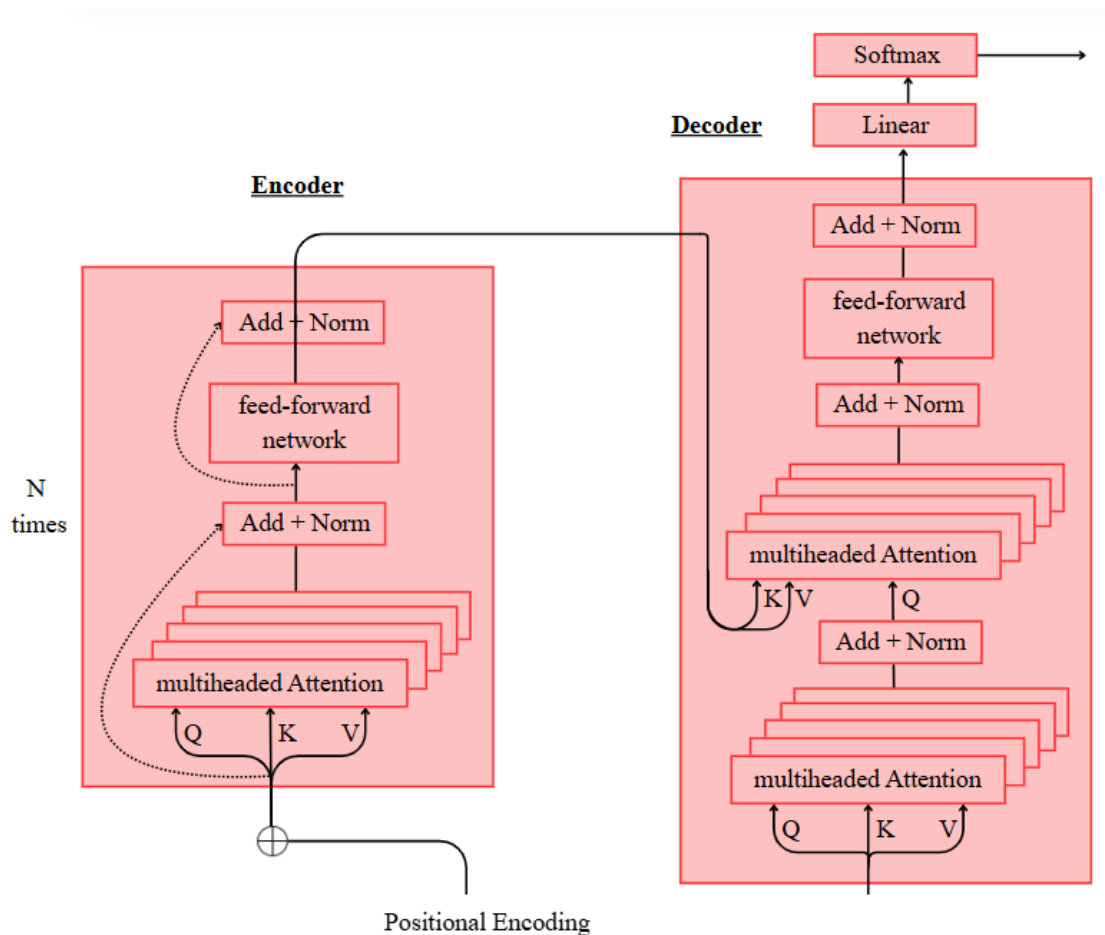
Example:

- $\rightarrow Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right) V$
- $\rightarrow Multihead(Q, K, V) = concat(head_1 head_2 \dots head_n) W_0$
- $\rightarrow head_i = Attention(W^Q_i Q, W^K_i K, W^V_i V)$



Note: "h" is the number of heads.

Transformer



Note: Sometimes the multiheaded attention is masked, i.e., it repeatedly pretends that the network had perfectly translated the first few words and hides the remaining words, to see if the network can predict subsequent words accurately.